

Henry Qi

916-693-0784 | henryqi@berkeley.edu | [linkedin.com/in/henry-qi-2026qi](https://www.linkedin.com/in/henry-qi-2026qi) | US Citizen

EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. in Electrical Engineering and Computer Science (EECS)

Aug. 2022 – May 2026

- GPA: 3.75/4.00
- Relevant Courses: Operating Systems, Database Systems, Efficient Algorithms, Machine Learning
- Swim Club: 4x National Championship Qualifier

EXPERIENCE

Software Engineering Intern

Jan 2026 – Present

CoPatible

Berkeley, CA

- Engineered a high-throughput Java backend pipeline, optimizing data processing workflows and improving system throughput and stream efficiency by 30% under production-scale workloads.
- Built scalable TypeScript/React dashboards to monitor real-time API performance, reducing incident response time by 35% and improving system reliability visibility across distributed services.
- Implemented secure multi-tenant data isolation architecture, enhancing data protection and reducing cross-tenant risk by 40% while accelerating regression testing workflows by 25%.

Teaching Assistant

Jan 2024 – Present

UC Berkeley Computer Science Department

Berkeley, CA

- Taught and mentored 50+ students in core electrical engineering concepts (Bode plots, RLC circuits, stability/controllability, PCA), improving average exam scores by ~20–30% through structured problem-solving.
- Designed and implemented standardized grading rubrics for exams and assignments, increasing grading consistency and reducing evaluation time by ~25% while better aligning assessments with learning objectives.
- Delivered targeted, iterative feedback loops that helped 80%+ of students improve performance, strengthening conceptual understanding and reducing common error rates in complex circuit/system analysis.

PROJECTS

AI Interview Evaluation Pipeline

- Evaluated 50+ interview transcripts (~400+ candidate responses) by converting raw conversations into structured JSON cases, scoring answers with Mistral against a rubric, and generating detailed per-candidate score reports.
- Designed a two-agent loop where Agent 1 scores and Agent 2 audits bias and structure, correcting roughly 15–20% of overly harsh ratings and confirming valid evaluator JSON in ~90% of runs.
- Migrated from a Gemini browser workflow to a Mistral JSON backend, achieving roughly 50–60× higher evaluation throughput (up to ~120–150 cases/minute on a single machine) with fewer JSON parsing errors.

AI Tutor Evaluation

- Evaluated ~1000 AI tutoring sessions with rubric-based scoring on engagement, scaffolding, and goal alignment, producing QA-style feedback reports that highlight strengths and gaps in tutor behavior.
- Added Redis-backed semantic memory and tracing so tutor sessions are stateful and fully observable, reducing time to diagnose tutor behavior issues from hours to minutes.
- Built a multi-model verdict engine where multiple LLMs independently score the same sessions, logging ~10–20% disagreement cases (~300+) in MongoDB Atlas as “hard examples” for model and prompt improvement.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, TypeScript, Rust, HTML, CSS

AI Systems & Evaluation: MCP, LLM application development, structured JSON outputs, multi-agent workflows, multi-model evaluation, rubric design, session memory, tracing/observability, OpenTelemetry, Arize

Frameworks & Libraries: React, Node.js, Flask, FastAPI, Spring Boot, JUnit, PyTest, Jest, Selenium, gRPC

Databases & Infrastructure: PostgreSQL, MySQL, MongoDB Atlas, Redis, SQLite, vector embeddings, semantic retrieval, caching

Cloud & DevOps: AWS, GCP, Docker, CI/CD, Linux, Git

Systems & APIs: REST APIs, WebSockets, RDBMS integration